

FundFlow Accountant — Diagnostic & Repair Manual

This manual is for **accountants and reconciliation officers** who investigate balance drift, reconciliation exceptions, bank clearance gaps, legacy migration issues, and ledger integrity on a FundFlow tenant.

Admin URL: <https://<your-fund-domain>/admin>

Primary workspaces: Finance → Reconciliation, Bank clearing, Reports; System → Audit & System.

Companion references: fund-flow-one-pager-accountants.md, fund-flow-dynamics-accountants.md, fund-flow-dynamics.md, fund-flow-reconciliation-and-scheduler.md.

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1. How FundFlow accounting is structured

1.1 Chart of accounts

Account	Role	Pool mirror
Master Bank	Imported statement lines	—
Master Cash	Pool cash on hand	Must equal Σ member cash
Master Fund	Pool equity	Must equal Σ member fund
Member Cash	Per-member wallet	Mirrored to master cash
Member Fund	Per-member equity	Mirrored to master fund
Loan sub-account	Principal per loan	Per loan
Master Fees	Fee income	Master only

1.2 Two ledgers, one economic event

Cash and **fund** move independently until an operation links them (e.g. contribution collection moves cash → fund).

1.3 Intent vs clearance

Stage	What it records
Ledger posting	Economic intent (deposit accepted, cash-out approved, disbursement)
Bank clearance	Match to imported statement line — updates linkage, not a second cash posting when done correctly

Reference flows: fund-flow-dynamics.md §2–5.

2. Reconciliation layers

FundFlow uses **three** complementary mechanisms:

```

flowchart LR
    subgraph realtime["Realtime (every Transaction)"]
        RT1["Journal balanced?"]
        RT2["Member drift"]
        RT3["Master pool drift"]
        RT4["Late-fee exemptions"]
    end

    subgraph snap["Audit snapshots"]
        S1["fund:reconcile --daily"]
        S2["fund:reconcile --monthly"]
        S3["ReconciliationSnapshot table"]
    end

    subgraph batch["Control batch"]
        B1["fund:nightly-reconciliation"]
        B2["Domain sweeps"]
        B3["Auto-resolve"]
        B4["BatchPostingGate"]
    end

    TXN["Transaction posted"] --> realtime
    CRON["Scheduler 06:00-08:00"] --> snap
    CRON --> batch
    batch --> EXC["ReconciliationException queue"]

```

Layer	Command / trigger	Output
Realtime	TransactionObserver → ReconciliationService::onTransactionPosted()	Immediate exceptions; audit log
Snapshots	fund:reconcile	reconciliation_snapshots — historical audit PDF
Nightly batch	fund:nightly-reconciliation	Full exception sweep + auto-resolve + halt gate

Important: Snapshots are for **audit history**. The **exception queue** is for **active remediation**.

3. Daily and monthly schedule

All commands run **per tenant** (`--tenants=slug` when run manually).

Time	Command	Accountant focus
06:00	<code>fund:assert-master-invariants</code>	Quick Σ master vs members
06:20	<code>fund:reconcile --daily</code>	Yesterday's audit snapshot
06:30	<code>fund:nightly-reconciliation</code>	Exception queue refreshed
07:15	<code>contributions:apply-late-fees</code>	Fee posting review
08:00	<code>bank:auto-match</code>	Unmatched lines reduced

Monthly: `fund:reconcile --monthly` on 2nd; contribution/EMI cycle jobs on 1st–6th (see `manual-administrator.md` §19).

Jobs UI: Audit & System → Jobs — manual re-run with history in `SystemJobRun`.

4. Investigation workflow

Standard loop for any reported balance issue:

1. Identify symptom (member complaint, exception badge, pool health widget)
2. Open Reconciliation → Queue (or member Transactions tab)
3. Read exception code + presenter guidance
4. Trace reference transaction(s) in audit log
5. Verify bank clearing state if cash-related
6. Apply correction (journal, reverse, reclassify, or CLI repair)
7. Re-run snapshot or wait for nightly batch
8. Confirm exception resolved and invariants pass

4.1 Data to collect before changing anything

Question	Where to look
Which member?	Member → Transactions (cash/fund tabs)
Which account type?	Cash vs fund vs loan
When did it start?	Activity date, migration date
Open exceptions?	Reconciliation → Queue
Uncleared bank lines?	Bank clearing → Work queue (operations)
Recent admin action?	Audit & System → Audit log
Stored balance vs txn sum?	<code>accounting:rebuild-balances --dry-run</code>

4.2 Do not skip dry-run

For balance rebuild and legacy repair commands, always run `--dry-run` first when available.

5. Pool invariants

5.1 Master pool

Service: `MasterAccountInvariantService`

CLI: `php artisan fund:assert-master-invariants --tenants=<slug>`

Check	Formula
Master cash	<code>master_cash.balance = Σ member_cash.balance</code>
Master fund	<code>master_fund.balance = Σ member_fund.balance</code>

Exception code	Severity
MASTER_CASH_POOL_DRIFT	high
MASTER_FUND_POOL_DRIFT	high
MASTER_IMBALANCE_UNRESOLVED	critical (halts batch posting)

Note: Realtime pool checks **skip** while `AccountingService::masterPoolMirrorInProgress()` — paired mirrors post master and member legs in one logical operation.

5.2 Tiny drift rounding

Nightly batch may post **rounding suspense** adjustments when drift is within configured tolerance (Settings → Reconciliation).

6. Member drift formulas

Service: `MemberInvariantService`

Code	Meaning
MEMBER_CASH_DRIFT	Stored cash component formula (deposits, disbursements, collections, repayments, cash-outs, etc.)
MEMBER_FUND_DRIFT	Stored fund component formula (contributions, loan fund legs, opening balances, guarantor debits, etc.)

Investigation:

1. Open member → **Transactions** → cash or fund tab.
2. Sum movements by type; compare to expected formula (see spec in workspace rules / fund-flow accountants doc).
3. If stored `accounts.balance` sum of `transactions` lines → run balance rebuild (§11).

Common causes: deleted transactions without balance adjustment; legacy import repair; manual SQL outside application.

7. Reconciliation exception playbook

UI: Reconciliation → **Queue**

Presenter copy: `ReconciliationExceptionPresenter` (titles and remediation hints in UI)

7.1 Master / journal

Code	Typical cause	First action
UNBALANCED_ENTRY DR CR on journal		Reverse or post balancing leg
MASTER_IMBALANCE_UNRESOLVED	members beyond tolerance	Fix root drift; clear halt

Code	Typical cause	First action
MASTER_*_POOL_DRIFT	Mirror missing or partial	Find txn without paired master leg
MEMBER_*_DRIFT	Component vs stored	Trace member ledger; rebuild balances

7.2 Contributions

Code	Typical cause	First action
PENDING_PAST_WINDOW_CLOSE	Uncollected still pending after cycle close	Collection ops — not always ledger bug
COLLECTED_WITHOUT_POST	Post collected, no txn	Post or reverse status
CONTRIBUTION_EXEMPTED_COLLECTED	Exempted ledger collected	Reverse collection
DUPLICATE_CONTRIBUTION_DEBIT	Duplicate debit	Reverse duplicate
ORPHAN_MASTER_FUNDING_CREDIT	Tag without member	Add member leg or reverse
CONTRIBUTION_MISSING_MASTER_CREDIT	Missing master credit	Post mirror
CONTRIBUTION_AMOUNT_MISMATCH	Amount mismatch	Adjust or reclassify

7.3 Loans / EMI

Code	Typical cause	First action
ACTIVE_BEFORE_FULL_DISBURSE	Loan disbursement	Fix loan status or post disbursement
DISBURSEMENT_MEMBER_CASH_MISSING	Disbursement without cash credit	Post cash leg
EMI_COLLECTED_LEDGER_MISSING	EMI paid, no txn	Post EMI collection
EMI_MISSED_SUFFICIENT_CASH	EMI paid but EMI not collected	Run collection or manual EMI post
EMI_OVER_COLLECTION	Paid more than installment	Reclassify excess
GUARANTOR_BORROWER_DUPLICATE_DEBIT	Duplicate debit	Reverse duplicate

7.4 Bank clearing

Code	Typical cause	First action
RECON_UNMATCHED_BANK_LINE	Bank line not matched	Bank clearing → match
RECON_AMBIGUOUS_MATCH	Multiple candidates	Manual match
UNMATCHED_CASH_ENTRY	Operational line stale	Match or clear
CASH_DEPOSIT_UNBANKED	Accepted deposit, no bank evidence	Import + match
AMOUNT_MISMATCH	Linked amounts differ	Adjust link or posting
STALE_PENDING	Old uncleared line	Review and match/ignore

7.5 Fees

Code	Typical cause	First action
FEE_WRONG_TIER	Late fee tier mismatch	Reapply correct tier
REPLACEMENT_PRIOR_FEE_NOT_REVERSED	Replacement fee not reversed	Reverse prior fee leg
FEE_INCOME_DRIFT	Master fees posted fees	Trace fee transactions

7.6 UI actions on exceptions

Available per code (see `ReconciliationExceptionActions`):

- **Retry auto-resolve**

- **Assign / Escalate / Write off / Accept override**
- **Custom journal / Post correction / Post cash correction**
- **Reverse transaction**
- **Reclassify** (contribution repayment)

8. Bank clearing diagnostics

Workspace: Finance → Bank clearing

8.1 Queue filters

Filter	Shows
All	Full queue
From bank file	Imported statement lines needing action
From operations	Uncleared deposit/cash-out placeholders

8.2 Expected lifecycle

Deposit accept → uncleared operational line → import CSV → match → cleared
 Cash-out accept → uncleared operational line → bank transfer → match → cleared
 Bank import only → mirror to cash → post to member → (optional) auto-collection

8.3 Common mistakes

Mistake	Symptom
Same deposit via Path A + Path B	Double cash credit
Clearance treated as new posting	Duplicate cash legs
Match wrong amount/date	AMOUNT_MISMATCH

Services: BankClearingQueueService, BankClearingMatchService, BankTransactionClearanceService

9. Ledger tools

9.1 Member ledger

Members → {member} → **Transactions** — cash, fund, loan tabs with running context.

9.2 Master accounts

URL: /admin/master-accounts — master cash, fund, bank, fees with transaction relation managers.

Manual adjustments (master only): Credit / Debit / Refund header actions on account transaction managers.

9.3 Audit trail

Audit & System → **Audit log** — filter by reconciliation, loans, admin overrides.

Export via Reports → audit export.

9.4 Imported loan repayments

Loans → {loan} → **Imported/legacy repayments** — historical migration repayment rows (paid at + amount). Compare sum to `master_portion` / `repaid_to_master` on loan.

10. Correction actions in the UI

Tool	Location	Use
Post correction entry	Exception row action	Paired journal per composer schema
Custom journal	Exception row action	Multi-leg manual entry
Reverse transaction	Exception or txn view	Undo incorrect posting
Reclassify	Exception action	Move amount contribution loan repayment
Master account manual CR/DR	Master account → transactions	Rare corrections (document reason)

Always: Log reason in exception notes; verify master pool after correction.

Prefer shared mirror helpers in `AccountingService` over one-off legs:

- `creditMemberCashWithMasterMirror` / `debitMemberCashWithMasterMirror`
- `creditMemberFundWithMasterMirror` / `debitMemberFundWithMasterMirror`

11. Artisan command runbook

Run from application server with tenant context:

```
php artisan <command> --tenants=<tenant-slug>
```

11.1 Reconciliation & balances

Command	Options	When to use
<code>fund:assert-master-invariants</code>		Quick pool check
<code>fund:reconcile</code>	<code>--no-store</code>	Point-in-time audit without saving
<code>--realtime</code>		
<code>fund:reconcile --daily</code>	—	Store daily snapshot
<code>fund:reconcile --monthly</code>	—	Month-end audit
<code>fund:nightly-reconciliation</code>	—	Full exception sweep
<code>accounting:rebuild-balances--dry-run</code>		Stored balance txn lines
<code>bank:auto-match</code>	—	Batch match attempt

Rebuild balances recalculates `accounts.balance` from sum of transaction lines — use after migration repair or deleted txn bugs. **Always dry-run first.**

11.2 Legacy migration repair

Command	Options	When to use
<code>legacy:repair-excess-loan-repayments--member=</code>		Repayments exceed fund-portion target
<code>legacy:repair-misclassified-contributionsdelinquent,</code> <code>--legacy-routed</code>		Contributions should be loan repayments
<code>legacy:repair-classified-paymentsmisclassified=</code>		Fix classified CSV before re-import
<code>legacy:rebuild-loan-installmentsloan=</code>		Schedule count/status wrong
<code>legacy:sync-loan-schedules --loan=</code>		Apply repayments to installments
<code>legacy:reimport-loan-repaymentspayments=, --dry-run</code>		Re-run repayment import
<code>legacy:migrate</code>	full options	Full migration (coordinate with admin)

Fund-portion rule: Legacy loan repayments should stop when `master_portion` (+ settlement) is satisfied — not at the old 50%+16% formula. Excess repair moves overflow back to contributions.

11.3 Statements

```
php artisan statements:generate --tenants=<slug> --period=YYYY-MM [--notify] [--member=]
```

12. Legacy migration repair

UI: Audit & System → Migration

12.1 Typical post-migration issues

Symptom	Likely cause	Repair path
Member cash too high	Misclassified loan repayments as contributions; phantom cash from bad reversal	<code>legacy:repair-misclassified-contributions;</code> <code>accounting:rebuild-balances</code>
Loan repayments > master portion	Old formula target (99k vs 79.5k)	<code>legacy:repair-excess-loan-repayments</code> <code>--loan=</code>
Installments wrong count	Schedule rebuild needed	<code>legacy:rebuild-loan-installments</code>
Pool drift after import	Missing mirrors or deleted txns	Rebuild balances + nightly recon

12.2 Safe repair order

1. Repair classification data (CSV or misclassified commands)
2. Repair loan repayment totals (excess command)
3. Sync loan schedules / rebuild installments
4. `accounting:rebuild-balances --dry-run` → then live
5. `fund:nightly-reconciliation`
6. `fund:assert-master-invariants`

12.3 Verification queries (conceptual)

For loan L after repair:

- $\Sigma \text{loan_repayments.amount} \text{ loan.fullRepaymentThreshold() (master_portion + settlement)}$
- `loan.repaid_to_master loan.master_portion`
- Member cash/fund pass `MemberInvariantService`

13. Year-end close gates

Audit & System → Year-end close

Readiness report checks (among others):

- Master pool mirrors
- Member drifts
- Open fund postings / cash-outs
- Open or critical reconciliation exceptions
- Uncleared bank lines
- Contribution cycle completeness
- Loan portfolio consistency

Do not close with open critical exceptions unless policy explicitly allows write-off.

See `fiscal-year-end-close.md`.

14. Reports and snapshots

Output	Where
Reconciliation PDF	Reports → reconciliation export; or snapshot row download
Collections CSV/XLSX	Reports
Loan portfolio	Reports
Audit log export	Reports
Snapshot history	Reconciliation → Snapshots tab

Snapshots include: ledger mismatch count, unposted bank rows, open exception count, coverage matrix by flow.

15. When batch posting is halted

Cause: `MASTER_IMBALANCE_UNRESOLVED` or manual halt in settings.

Effect: Halt-sensitive jobs blocked (contribution apply, EMI apply, late fees, bank auto-match, etc.) — see `ScheduledJobRegistry` `halt_sensitive: true` entries.

Recovery:

1. Fix underlying master/member drift.
2. Resolve or write off critical exceptions.
3. Run `fund:nightly-reconciliation` — clears gate on success.
4. Or admin clears halt in settings after verification.

Jobs UI shows halt reason when present.

16. Symptom → entry point quick table

Symptom	Start here
Reconciliation badge	Finance → Reconciliation → Queue

Symptom	Start here
Bank queue badge	Finance → Bank clearing
Member says balance wrong	Member → Transactions + <code>accounting:rebuild-balances --dry-run</code>
Master sum of members	<code>fund:assert-master-invariants</code>
Deposit accepted but not in bank	<code>CASH_DEPOSIT_UNBANKED</code> / bank queue operations
Double contribution	<code>DUPLICATE_CONTRIBUTION_DEBIT</code>
Loan shows wrong repayment total	Loan → Imported/legacy repayments; <code>legacy:repair-excess-loan-repayments</code>
Post-migration pool drift	\$12 repair order
Year-end blocked	Audit & System → Year-end close readiness
Need audit trail for auditor	Reconciliation → Snapshots + Reports

Appendix A — Standard journal patterns

Contribution collection

DR	CR
Member cash	
Master cash (mirror)	
	Member fund
	Master fund (mirror)

Loan disbursement

DR	CR
Member fund (portions)	
Master fund (mirror)	
Loan account	
	Member cash
	Master cash (mirror)

Cash-out approval

DR	CR
Member cash	
Master cash (mirror)	

Plus uncleared `bank_transactions` row.

Loan repayment

1. Cash-in mirror (CR member + master cash)
2. Cash debit (DR member + master cash)

3. Fund credit (CR member + master fund)
4. Loan principal credit (CR loan account)

Appendix B — Related documentation

Document	Topic
fund-flow-dynamics.md	Full money-flow diagrams
collection_cycle_workflow.md	Collection phases
loan-repayment-operations.md	EMI and repayment
production-runbook.md	Server operations
.cursor/rules/accounting-master-member-sync.mdc	Mirror rules for developers

Appendix C — Tests as behavioural spec

When in doubt about expected behaviour, consult:

Test file	Covers
<code>tests/Feature/Tenant/ReconciliationAndMigrationClosureTest.php</code>	EMI and repayments, pool drift
<code>tests/Feature/Tenant/BankClearingQueueWorkflowTest.php</code>	Clearing workflow
<code>tests/Feature/Tenant/LegacyMigrationTest.php</code>	Legacy repair scenarios
<code>tests/Feature/Tenant/FiscalClosePhaseThreeTest.php</code>	Year-end gates
<code>tests/Feature/Tenant/ComplianceLayerTest.php</code>	Invariant compliance
